



[REFERENCE IMPLEMENTATION]

CLINICAL RISK PREDICTION · ICU EARLY-WARNING SYSTEM · PORTFOLIO

Sepsis Early-Warning Risk Analytics

An XGBoost early-warning classifier over ICU vitals and labs — excellent at ranking risk, and honestly reported on the one place its raw probabilities should not yet be trusted.

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Gradient-boosted early warning · SHAP explainability · calibration-audited risk scoring

A model that ranks deterioration risk excellently — and a probability number that overstates itself

This project predicts sepsis onset within 24 hours from 6-hourly ICU vitals and labs across 6,500 stays. Discrimination is strong: the model separates high-risk windows from low-risk ones with 0.879 AUC-ROC. Its raw probability outputs, however, are measurably overconfident — reported here rather than smoothed over.



0.879

Test AUC-ROC — strong
discrimination between risk levels



76.8%

Sensitivity at 90% specificity —
the evaluated operating point



44.1%

Precision at that operating point —
1.3 false alerts per true catch



Overconfident

Predicted risk runs well above
observed risk in every decile



Bottom line

This is a research reference implementation for triage support, not a diagnostic device — every figure here is scoped to a synthetic cohort and requires clinical validation before real use.

Presentation agenda



01

The Clinical Problem

Why sepsis recognition is a race against hours



02

Data & Cohort Profile

6,500 stays, 39,034 six-hourly windows



03

Feature Engineering

Twenty-five features, and which ones actually matter



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Modeling: Gradient-Boosted Risk

An XGBoost classifier, tuned for rare events



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Evaluation & Validation

Strong ranking, and a calibration problem worth naming



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Clinical Impact

What a flag means for a 6,500-stay ICU, honestly



07

Deployment & Responsible Use

Dashboard, model card, and clinical-AI guardrails

01 — 07



SECTION ONE

The Clinical Problem

Sepsis mortality rises with every hour treatment is delayed — the earliest reliable warning is the one that matters most.

SECTION 1 OF 7



THE STAKES

A time-critical diagnosis, hiding in noisy signals

Sepsis is a life-threatening response to infection that can progress from subtle to critical within hours — and its earliest signs overlap with dozens of less urgent conditions common in an ICU.



A real base rate

20.3% of stays in this cohort develop sepsis at some point — common enough that a structured early-warning signal has real room to help.



A narrow detection window

The model's task is deliberately tight: is sepsis likely in the next 24 hours, from the vitals and labs available right now.



A high, but informative, bar

0.879 AUC-ROC on held-out data — the level of discrimination every downstream claim in this deck is measured against.

The question this project asks: can a structured, repeated 6-hourly risk score surface early deterioration a clinician might not yet have connected into a pattern?

Criteria-based screening, checked intermittently

Established screening tools exist — but they are typically applied at intervals, and rely on a clinician synthesizing many data points under real time pressure.



Manual criteria review

Screening tools require a clinician to pull together several vitals and labs and check them against threshold criteria, by hand



Checked, not continuous

Review typically happens on rounds or at fixed intervals — not continuously as every new vital and lab result arrives



Twenty-five inputs, one score

This project’s model folds vitals, labs, and their trends into a single number, recomputed every 6-hour window automatically



This project’s target

A structured risk score at every window — feeding clinical judgment with a consistent signal, not replacing the decision

6,500

ICU stays in
this cohort

This is the fleet-wide cohort every rate and probability in this deck is computed against — 39,034 six-hourly windows in total.

A repeated risk score, not a one-time diagnosis

This model never diagnoses sepsis — sepsis is diagnosed clinically, by a physician. What it does is re-estimate, every 6 hours, how likely deterioration is in the next 24, from the data already being collected.

What the model outputs

**P(sepsis onset
within 24h)**

One number per window, recomputed as new vitals and labs arrive — not a single admission-time prediction.

What happens with that number



Above threshold: flagged as elevated risk for clinical review



Below threshold: no flag — standard monitoring continues



Always reviewed by a clinician — never an autonomous treatment trigger

What this project set out to prove



Predict deterioration early

Estimate sepsis-onset risk 24 hours ahead, from data already collected at the bedside.



Achieve strong discrimination

Reliably separate high-risk windows from low-risk ones across a held-out test set.



Identify the real risk drivers

Determine which vitals, labs, and trends actually carry predictive weight — not just correlate loosely.



Audit the model's own confidence

Check whether predicted probabilities match observed outcomes — not just whether rankings are correct.



Validate on a genuine held-out set

Report test-set performance separately from training and validation, at a stated operating point.



Scope this as research, explicitly

Package a model card and guardrails that keep this a reference implementation, not a deployed device.



SECTION TWO

Data & Cohort Profile

6,500 ICU stays, two ways to count how many patients develop sepsis, and a clear physiological signal underneath both.



DATASET

One cohort, six-hourly windows, one held-out test set

Every ICU stay contributes a sequence of 6-hourly windows, each carrying that moment’s vitals, labs, and a label for whether sepsis onset followed within 24 hours.

Source files

icu_stays.csv	39,034 windows — 6,500 stays, vitals + labs	Cohort
sepsis_xgb.joblib	fitted XGBoost model + feature list	Model
test_idx.npy	row indices of the 5,849-window test set	Eval
metrics.json	AUC-ROC, split sizes, positive rates	Eval
evaluation_report.json	sensitivity, specificity, Brier score	Eval

app.py	clinician dashboard — per-stay risk + SHAP	App
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Cohort at a glance

6,500

ICU stays

39,034

six-hourly windows

20.3%

stays that develop sepsis

9.0%

windows labeled positive

0.879

test-set AUC-ROC

KEY INSIGHT

20.3% and 9.0% are both correct — they answer different questions

“20.3% of stays develop sepsis at some point (1,318 of 6,500 patients). But only 9.0% of individual 6-hourly windows are labeled positive — because most of a septic patient’s stay is still pre-onset. The model is trained and evaluated on the window-level number.”



Patient-level: 20.3%

The share of ICU stays where sepsis develops at any point — the number that matters for staffing and program planning.



Window-level: 9.0%

The share of individual 6-hour observations labeled positive — the number the classifier actually trains and is scored against.



Why both belong in this deck

Quoting only one figure invites a reader to misjudge either how common sepsis is, or how rare a positive window is for the model to catch.

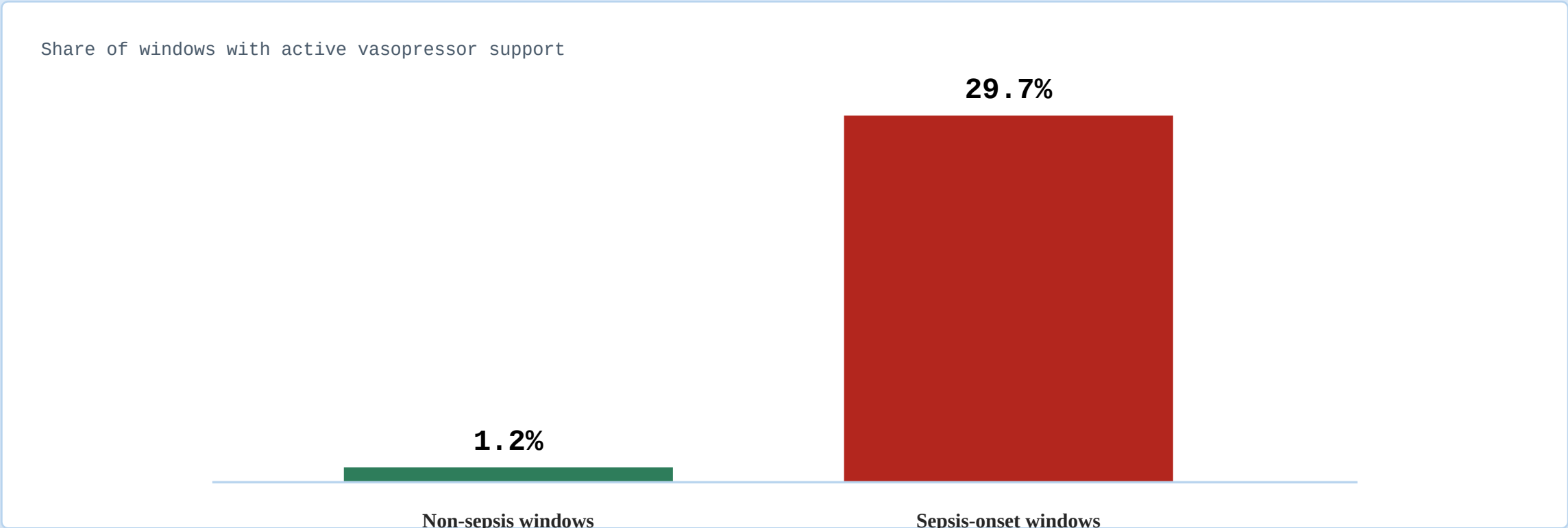
Vitals and labs diverge clearly before onset

Mean values in the 24 hours before sepsis onset, compared to windows that stay onset-free — a pattern consistent with the physiology sepsis screening tools are built around.

Signal	Non-onset	Sepsis-onset	Change
Lactate (mmol/L)	1.37	2.61	+90%
WBC (×10 ³ /μL)	8.45	11.73	+39%
Heart rate (bpm)	83.4	93.4	+12%
Resp. rate (/min)	17.4	20.6	+18%
MAP (mmHg)	77	69.8	−9%

Vasopressor use, non-onset vs. sepsis-onset windows

Vasopressor support — medication to sustain blood pressure — is part of the clinical definition of septic shock, so this gap is expected. It is also, unsurprisingly, one of the model’s strongest features (Slide 19).



Baseline demographics, almost identical across groups



Age

62.9 vs. 63.0 years — no meaningful difference between windows that precede onset and those that don't.



Comorbidity index

2.22 vs. 2.24 — patients heading toward sepsis are not, on average, carrying a heavier baseline disease burden.



Frailty index

0.505 vs. 0.486 — essentially flat, reinforcing that acute physiology, not baseline frailty, is doing the predictive work.

This is a finding, not a null result to gloss over: it means the model earns its predictive power from what is happening physiologically right now, not from who the patient was on admission.

A synthetic ICU cohort, disclosed as such



Why synthetic

A generated cohort removes any dependence on identifiable patient records for a portfolio piece, while preserving realistic sepsis physiology.



What was preserved

A genuine, checkable separation in lactate, WBC, and vasopressor use between onset and non-onset windows, consistent with real sepsis criteria.



What is disclosed

No stay here corresponds to a real patient. Every downstream number is scoped to this synthetic cohort until validated on real ICU telemetry.



A methodological choice, stated up front

No claim in this deck should be read as validated on real patient data — that validation is named explicitly as a precondition for any clinical use (Slide 38).



SECTION THREE

Feature Engineering

Twenty-five vitals, labs, and trends feed the model — but not all of them, or even most of them, carry equal weight.

THE FEATURE SET

Level, trend, and patient context — twenty-five in all

Every window carries the current value of ten signals, the change since the prior window for those same ten signals, and five patient-context features.

Ten vitals & labs, twice (20 features)

heart_rate	heart_rate_delta
map_bp	map_bp_delta
resp_rate	resp_rate_delta
temp_c	temp_c_delta
spo2	spo2_delta
lactate	lactate_delta
wbc	wbc_delta
creatinine	creatinine_delta
platelets	platelets_delta
fluid_balance_ml	fluid_balance_ml_delta

Patient context (5 features)

age	patient age in years
comorbidity_index	count of pre-existing conditions
frailty_index	a standardized frailty score
vasopressor_flag	active blood-pressure support, yes/no
hours_since_admission	how far into the stay this window falls

KEY INSIGHT

For sepsis, where you are matters more than how fast you got there

“The current value of lactate carries roughly 10× the predictive weight of its own trend. Every _delta feature ranks below every one of the top clinical severity markers — the opposite pattern from a slow-degrading mechanical system.”



A clinically grounded result

Lactate and vasopressor need are established, real-time severity markers in sepsis criteria — the acute state itself carries diagnostic weight, not just its slope.



Different from a bearing, on purpose

A mechanical part fails from accumulated wear, so its trend dominates. A patient’s crisis is defined by their current physiology — this is a genuinely different failure mode.



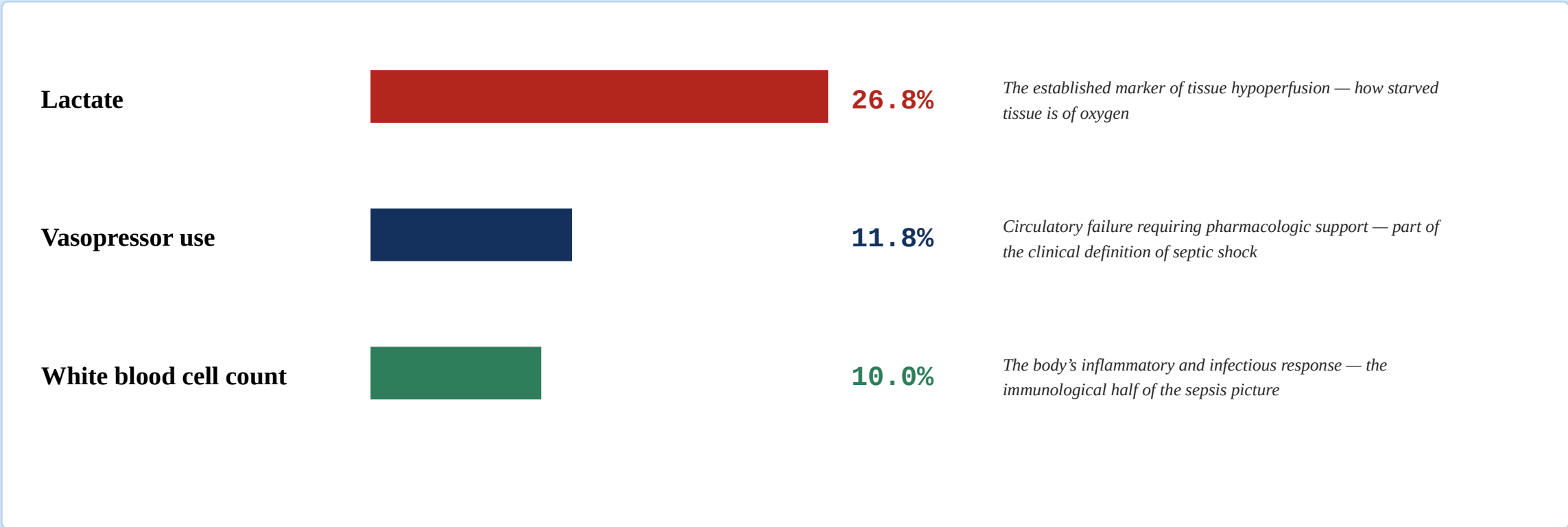
Trend features still matter, modestly

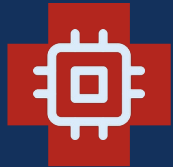
Every delta feature carries real, non-zero weight — they are not noise, just consistently smaller contributors than the level they describe.

THREE FEATURES, HALF THE MODEL

Lactate, vasopressor use, and white cell count

These three features alone account for 48.6% of the model’s total decision weight — and all three map directly onto the core clinical definition of sepsis and septic shock.





SECTION FOUR

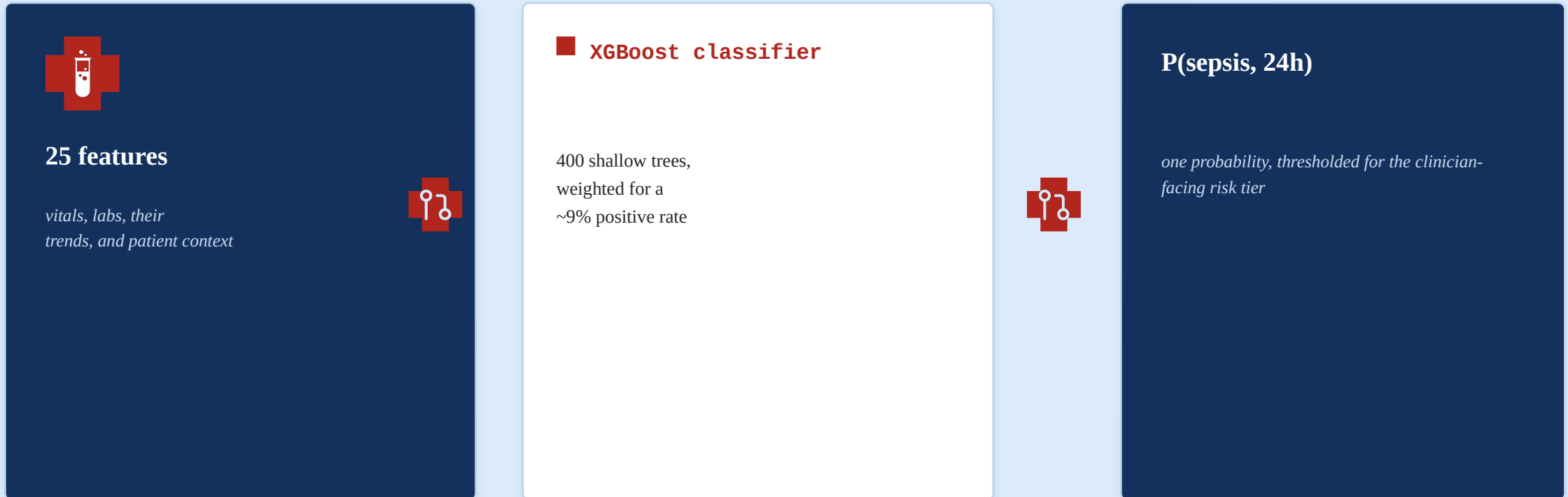
Modeling: Gradient-Boosted Risk

An XGBoost classifier tuned for a rare, high-stakes event, trained on twenty-five clinically grounded features.



Features in, one probability out, every six hours

One gradient-boosted classifier handles the entire task — twenty-five features per window produce a single, recomputed sepsis-risk probability.



Shallow, regularized trees, weighted for a rare event

Every hyperparameter leans toward avoiding overfitting on a modest cohort, while scale_pos_weight compensates for sepsis-onset windows being a clear minority.

n_estimators 400 Boosting rounds	subsample 0.85 Row sampling per tree
learning_rate 0.05 Conservative shrinkage	eval_metric auc Optimized directly for ranking quality
max_depth 4 Shallow trees — controls overfitting	colsample_bytree 0.85 Column sampling per tree
scale_pos_weight 10.36 Reweights the minority positive class	random_state 42 Reproducibility

Reweighting, not resampling

At roughly 9% window-level prevalence, sepsis-onset windows are a real minority — addressed with a loss-function weight, not by inventing or discarding rows.



scale_pos_weight ≈ 10.36

Every sepsis-onset window's training loss is scaled up roughly tenfold — no synthetic windows, no discarded data.



Every training window stays real

Unlike oversampling techniques, this never invents a synthetic patient-window — every example the model sees is a genuine observation.



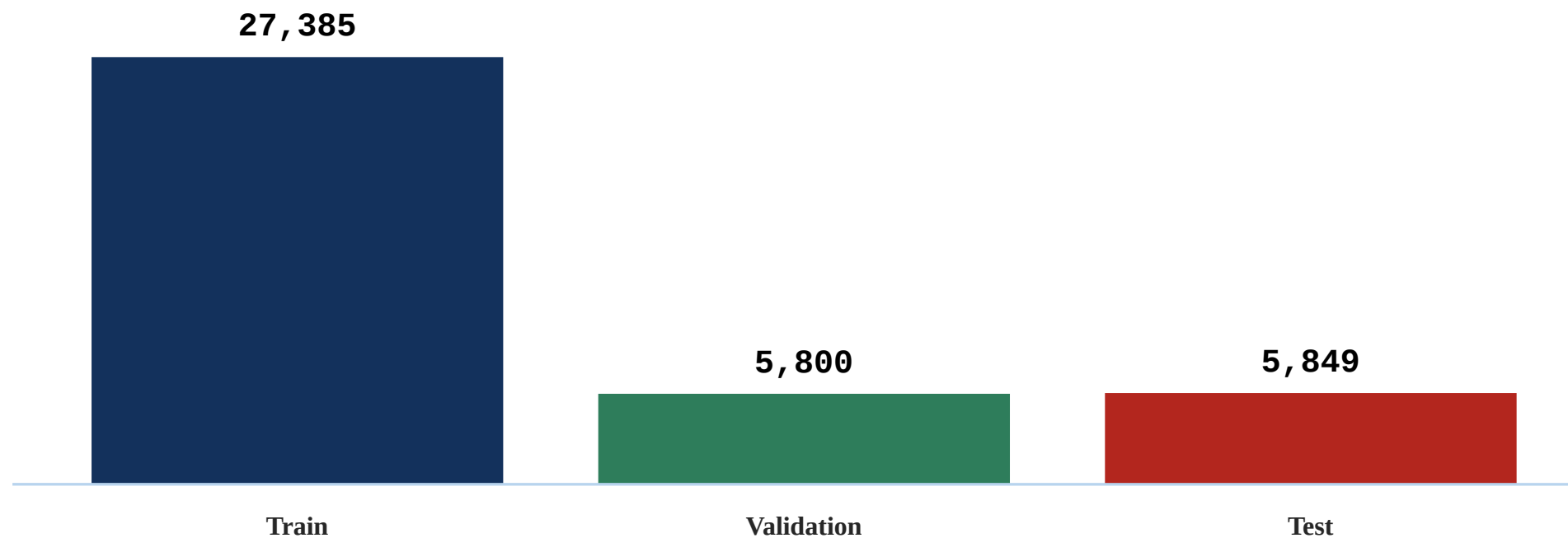
Why this matters clinically

Manufacturing synthetic minority-class examples in a clinical dataset risks encoding artifacts that don't reflect real physiology — reweighting avoids that risk entirely.

A three-way split, held apart before any tuning

Windows by split

Val AUC 0.8795, test AUC 0.879 — nearly identical, indicating the reported test performance is not a lucky split.

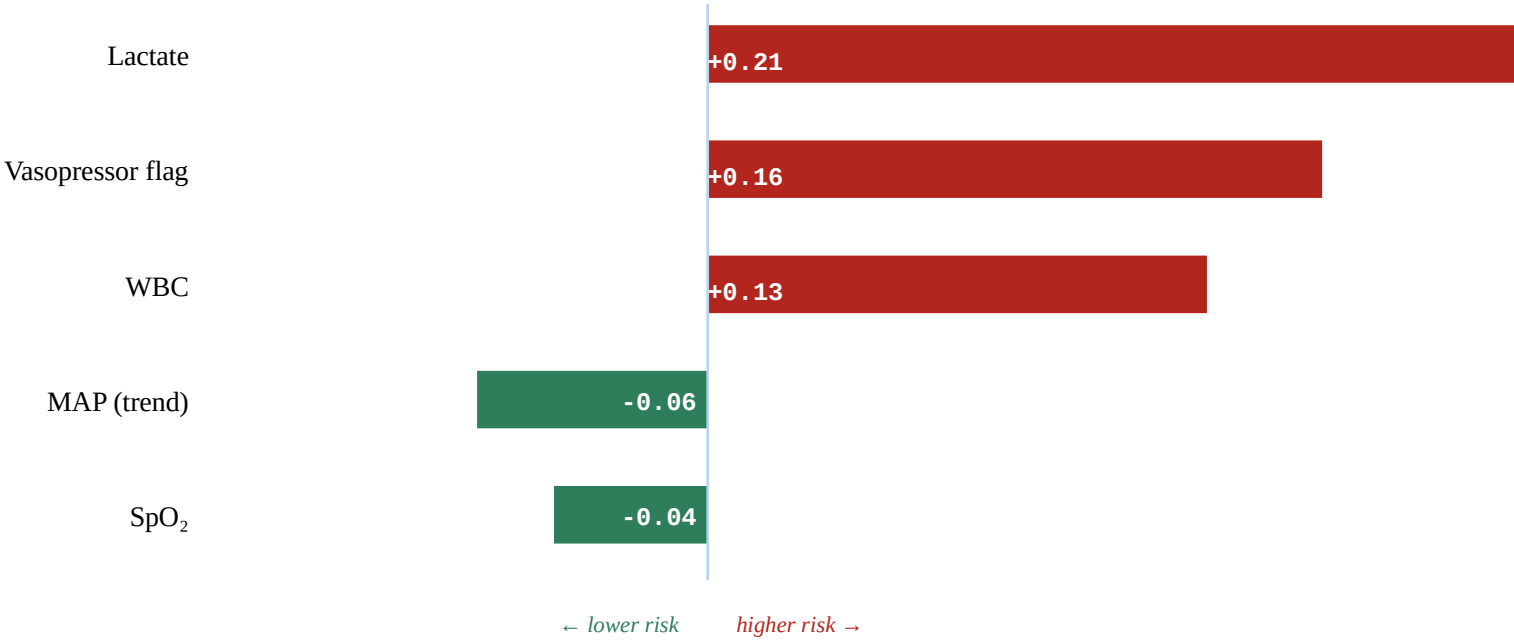


EXPLAINING EVERY PREDICTION

SHAP values, computed per patient, at the bedside

The dashboard doesn't just show a risk percentage — it shows which vitals and labs pushed that specific window's score up or down, using SHAP values computed live from the fitted model.

Typical direction of effect, by feature (illustrative single-window example)





SECTION FIVE

Evaluation & Validation

Strong discrimination on held-out data — and a calibration problem this deck names rather than smooths over.



Three questions, on 5,849 held-out windows

Discrimination, operating-point trade-offs, and calibration test different things — a model can excel at one and fall short on another.



Does it rank risk correctly?

AUC-ROC — across every possible threshold, do higher-risk windows score higher than lower-risk ones?



What happens at the chosen threshold?

Sensitivity, specificity, and the resulting confusion matrix at the operating point actually used.



Are the probabilities themselves trustworthy?

Brier score and a decile calibration check — does "70% risk" mean what it says?

Discrimination and the chosen operating point

0.879

Test AUC-ROC

0.50 = random guessing. 1.00 = perfect ranking, every time.

0.879 is a strong result for a rare-event clinical classifier on held-out data — comfortably above the 0.8 threshold commonly treated as "good" discrimination.

Operating point (threshold 0.4727)

76.8%

sensitivity — share of true sepsis-onset windows correctly flagged

90.0%

specificity — share of non-onset windows correctly cleared

This threshold was chosen to hold specificity near 90% — a deliberate trade-off toward limiting false alerts (Slide 30).

THE HONEST RESULT

Great at ranking risk. Not yet trustworthy as a literal probability.

“The model’s Brier score (0.0974) is worse than simply predicting the base rate for every window (0.0842). In the highest-risk decile, the model predicts 89% risk on average — the true observed rate is 62%.”



Discrimination and calibration are different things

AUC asks "does it rank correctly?" Brier score asks "do the numbers mean what they say?" This model excels at the first and falls short on the second.



Systematically overconfident, not random noise

Every decile shows predicted risk above observed risk — a consistent, correctable bias, not scattered error.



A known, fixable gap

Post-hoc calibration (Platt scaling or isotonic regression) is a standard fix — named explicitly as required before any literal probability is shown to a clinician (Slide 38).

What 946 flagged windows actually contain

Test set, n = 5,849 windows, threshold 0.4727

True positive 417 <i>sepsis-onset windows correctly flagged</i>	False positive 529 <i>non-onset windows flagged unnecessarily</i>
False negative 126 <i>sepsis-onset windows missed</i>	True negative 4,777 <i>non-onset windows correctly cleared</i>



SECTION SIX

Clinical Impact

What a flag means for a 6,500-stay ICU — and what the model's own numbers say about trusting it.



A prompt for review, not an automated action

The consequence of a false alert matters as much as the number itself — here, the response to a flag is deliberately conservative.



Elevated risk tier, on the dashboard

A flagged window surfaces as "elevated" or "high" risk with its contributing factors — visible to the care team, not sent anywhere automatically.



A prompt to look closer, not to act

The appropriate response to a flag is clinical review of the underlying vitals and labs — exactly what a bedside clinician would want to see anyway.



Why a 44% precision is workable here

A false alert costs a clinician a few minutes of chart review, not an invasive intervention — the asymmetry that makes a 1.3:1 alert-to-catch ratio reasonable.

This is a description of the tool's intended workflow, not medical advice — clinical protocols are set by the care team, always.

Scaling the test-set confusion matrix to the full cohort

Applying the same operating point across the entire 39,034-window cohort gives a concrete sense of daily alert volume — not a guarantee of real-world performance, but a useful order-of-magnitude check.



~16%

of all windows would be flagged for review at this threshold — 946 of 5,849 in the test set alone.



~1 in 3

flagged windows go on to develop sepsis within 24 hours — the other two are still worth the look, given the low cost of checking.



~1 in 4

true sepsis-onset windows are still missed at this threshold — a real, named gap this tool does not claim to close alone.

None of this replaces a prospective clinical trial — it translates a held-out test-set result into the scale a care team would actually experience, so the trade-off is concrete rather than abstract.

A strong ceiling, stated plainly



What it tells us

The model ranks sepsis-bound windows above non-onset windows 87.9% of the time for a random pair — a real, strong, stable signal across three separate splits.



What it doesn't tell us

It says nothing about individual diagnosis, nothing about causation, and — as Slide 29 shows — its raw probability output should not yet be read literally.



A dashboard worth double-checking anyway

This deck's reference dashboard's headline figures matched the authoritative model files on every number checked — a reassuring, but not sufficient, consistency check.

An honest ceiling, stated once, is worth more than an impressive-looking number that quietly doesn't hold up under a second look.



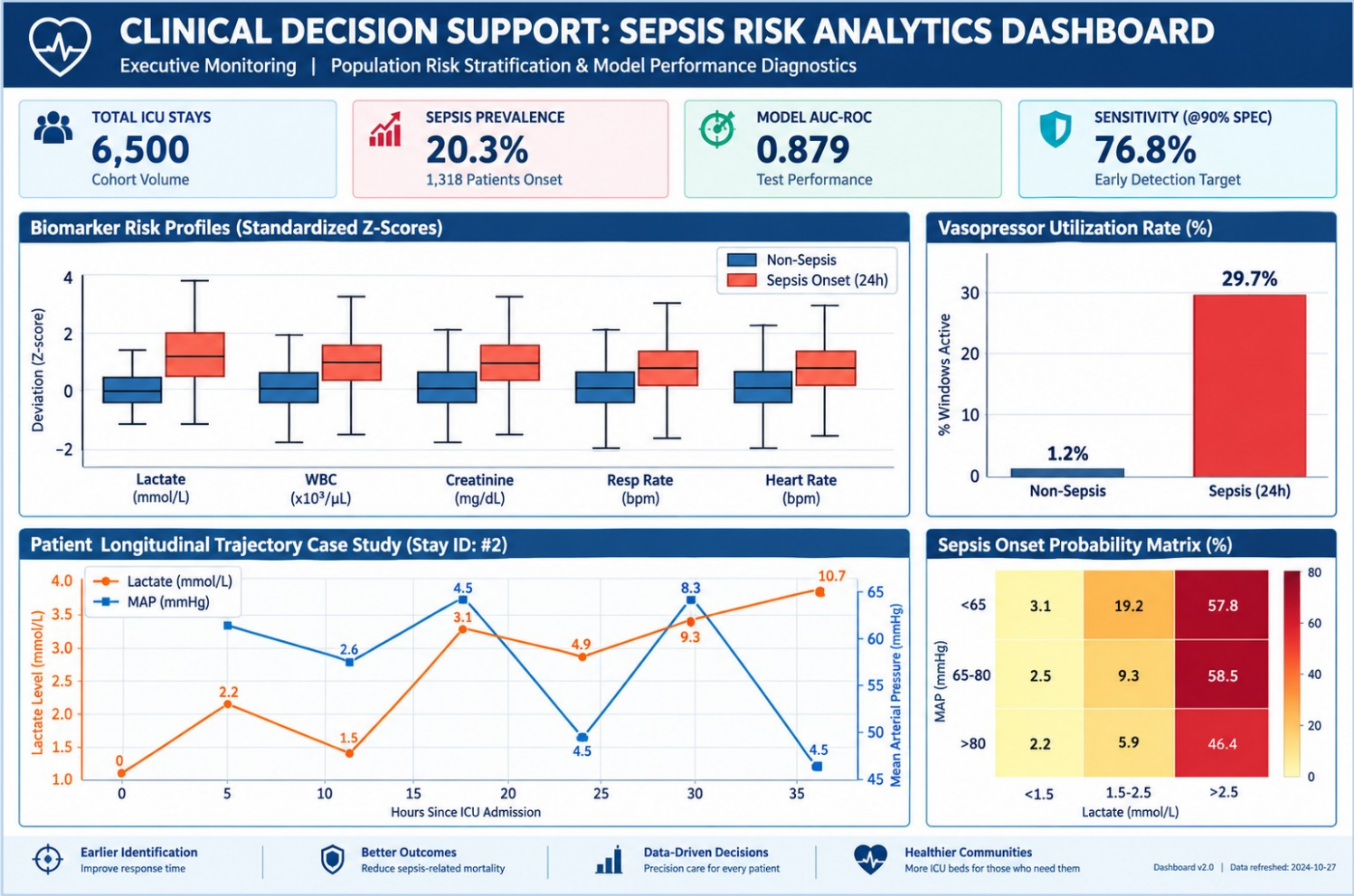
SECTION SEVEN

Deployment & Responsible Use

A dashboard, a model card, and the guardrails a clinical early-warning system needs before it earns a place in a real ICU.



The clinician dashboard, at a glance



What it surfaces

- Cohort volume, prevalence, model AUC & sensitivity
- Biomarker risk profiles by outcome class
- Vasopressor utilization by class
- Patient longitudinal trajectory case study
- Sepsis onset probability by MAP × lactate

Every headline figure on this dashboard was checked directly against the model files and matched — AUC, sensitivity, and vasopressor rates all confirmed exactly.

Model card summary



Intended use

A repeated, 6-hourly early-warning score to prompt clinical review of sepsis risk. Not a diagnostic tool; not validated on real patient data; always reviewed by a clinician.



Limitations

AUC 0.879 on a synthetic cohort; raw probabilities are measurably overconfident (Slide 29) and require calibration before literal display. 23.2% of true onset windows are missed at the evaluated threshold.



Next steps

Apply post-hoc probability calibration, validate on a real, IRB-approved ICU dataset, and pilot in shadow mode before it informs a real care decision (Slide 39).

Guardrails for a tool that touches patient care



A screening aid, never an autonomous decision

Every flag is reviewed by a clinician — the model surfaces a risk tier and its drivers; a person decides what happens next, every time.



Calibration is fixed before probabilities are shown literally

Until post-hoc calibration is applied, the dashboard should present a risk tier (low/elevated/high), not a raw percentage a clinician might over-trust.



No deployment without real-data validation

Every result in this deck is scoped to a synthetic cohort. IRB-approved validation on real, de-identified ICU data is a precondition, not a formality.



Miss rate reported alongside every win

The 23.2% of true onset windows missed at the operating threshold ships with the AUC and sensitivity numbers, every time — never presented in isolation.

From reference implementation to a validated tool



Calibrate the probability output

Apply Platt scaling or isotonic regression on a held-out calibration set, then re-check the decile analysis until predicted and observed rates align.



Real-cohort validation

Re-train and re-evaluate on an IRB-approved clinical dataset before any pilot use — every number in this deck resets at that point.



Threshold recalibration with clinicians

Set the operating threshold jointly with the care team, weighing sensitivity against alert fatigue — not a single fixed specificity target.



Prospective shadow-mode trial

Run the score in parallel with usual care, visible but non-actionable, before it influences a single real clinical decision.

KEY TAKEAWAYS

Strong risk ranking, and a calibration gap reported rather than hidden



0.879 test AUC-ROC



76.8% sensitivity @ 90% specificity



Overconfident probabilities, disclosed



Lactate + vasopressor + WBC = 49%

Skills demonstrated

Clinical risk modeling · gradient-boosted classification · SHAP explainability · calibration auditing · responsible clinical-AI reporting

Anthony Chinedu Echem

Machine Learning Engineer & Data Scientist



Portfolio contact available on request

